

KALASALINGAM
ACADEMY OF RESEARCH AND EDUCATION
(DEEMED TO BE UNIVERSITY)
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DIET AND NUTRITION ADVISOR

Project ID : PRJ022

Team ID : T022

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Project Trainer / Guide

Mr. SHIRWIN PRINCE I

I. INTRODUCTION

Healthify is an AI-powered web application designed to help users manage their daily diet, nutrition, and fitness activities. The platform provides personalized diet plans, food recognition, calorie tracking, workout guidance, and health progress monitoring.

The application uses Artificial Intelligence and Machine Learning-based technologies to analyze food images and identify nutritional information. It integrates AI-powered image recognition to detect food items and estimate their nutritional values, including calories, proteins, carbohydrates, and fats.

The system calculates the user's Basal Metabolic Rate (BMR) and Total Daily Energy Expenditure (TDEE) based on personal details such as age, height, weight, gender, and activity level. Based on these calculations, the application generates personalized diet plans consisting of five meals per day.

Users can track their daily food consumption, water intake, step count, and workout activities. The platform also provides a 25-day fitness challenge to encourage users to maintain consistency in their fitness routines.

Healthify uses a MySQL database to store user information, diet plans, food logs, and fitness activities. It also supports offline data access through IndexedDB and synchronizes stored information when the internet connection becomes available.

The main objective of this project is to provide an intelligent and user-friendly platform that helps users maintain healthy eating habits and monitor their fitness activities.

II. PROBLEM STATEMENT & OBJECTIVES

PROBLEM:

Maintaining a healthy lifestyle requires proper diet planning, nutrition monitoring, and regular physical activity. However, many people find it difficult to calculate their daily calorie requirements, identify nutritional values in food, prepare balanced diet plans, and consistently track their fitness activities.

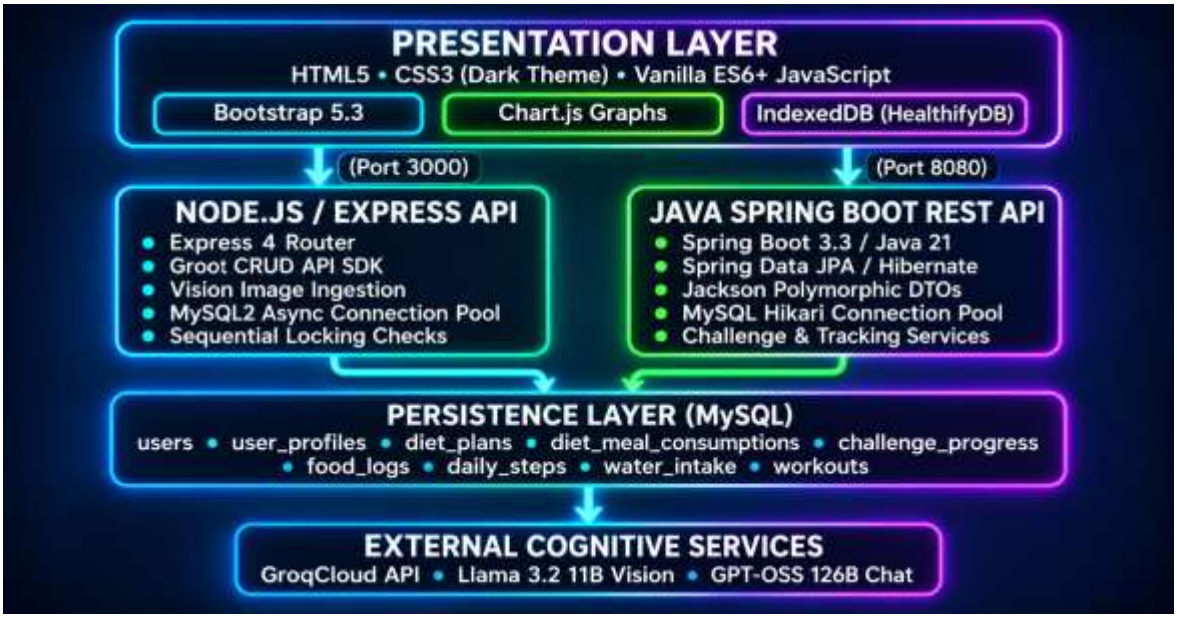
Traditional fitness applications may require users to manually enter food details and calculate nutritional information. This process can be time-consuming and may reduce user engagement.

To overcome these challenges, Healthify provides an AI-powered web application that helps users identify food items, understand nutritional information, generate personalized diet plans, and monitor their daily fitness activities in one platform.

OBJECTIVES:

- To develop a user-friendly web application for diet and fitness management.
- To calculate BMR and TDEE based on user profile information.
- To generate personalized diet plans based on calorie requirements.
- To identify food items from uploaded images using AI.
- To provide nutritional information such as calories, proteins, carbohydrates, and fats.
- To support prescription and document text extraction using AI-based OCR.
- To allow users to track daily food consumption and calorie intake.
- To monitor water intake, daily steps, and workout activities.
- To implement a 25-day fitness challenge with progress tracking.
- To provide workout guidance and supplement information.
- To store user information and activity records securely.
- To support offline access and automatic data synchronization.

III. SYSTEM ARCHITECTURE





Healthify follows a multi-layered architecture that integrates a web frontend, backend services, AI services, and a relational database. The system uses Node.js and Express as an AI gateway and Java Spring Boot for backend application services.

The application communicates with GroqCloud AI models for food image recognition, document processing, and AI-based responses. MySQL stores user information, diet plans, and fitness tracking records.

Working Process

1. **User Registration and Login:** Users create an account and log in to access the Healthify platform.
2. **Profile Management:** The user enters personal information such as age, gender, height, weight, and activity level.
3. **BMR and TDEE Calculation:** The system calculates the user's daily calorie requirements using the Mifflin-St Jeor equation and activity multiplier.
4. **AI Food Recognition:** Users upload food images, and the AI vision model identifies the food items and estimates their nutritional information.
5. **Nutrition Information Retrieval:** The system uses nutritional reference data from ICMR-NIN Indian Food Composition Tables and USDA FoodData Central to retrieve relevant food information.
6. **Diet Plan Generation:** Based on the user's calorie requirements and preferences, the system generates a personalized diet plan containing five meals per day.
7. **Daily Activity Tracking:** Users record their food consumption, water intake, daily steps, and workout activities.
8. **Fitness Challenge:** The system provides a 25-day fitness challenge in which users can complete daily activities and monitor their progress.

9. **Database Storage:** User profiles, diet plans, food logs, and fitness activities are stored in the MySQL database.
10. **Offline Data Synchronization:** IndexedDB stores selected information locally when offline and supports synchronization when the internet connection is restored.
11. **Dashboard and Progress Monitoring:** The application displays daily nutrition details, calorie intake, fitness activities, and user progress.

IV. TECH STACK



V. IMPLEMENTATION DETAILS

Figure 1: User Registration and Account Creation Interface



Figure 2: Health Tracker Dashboard

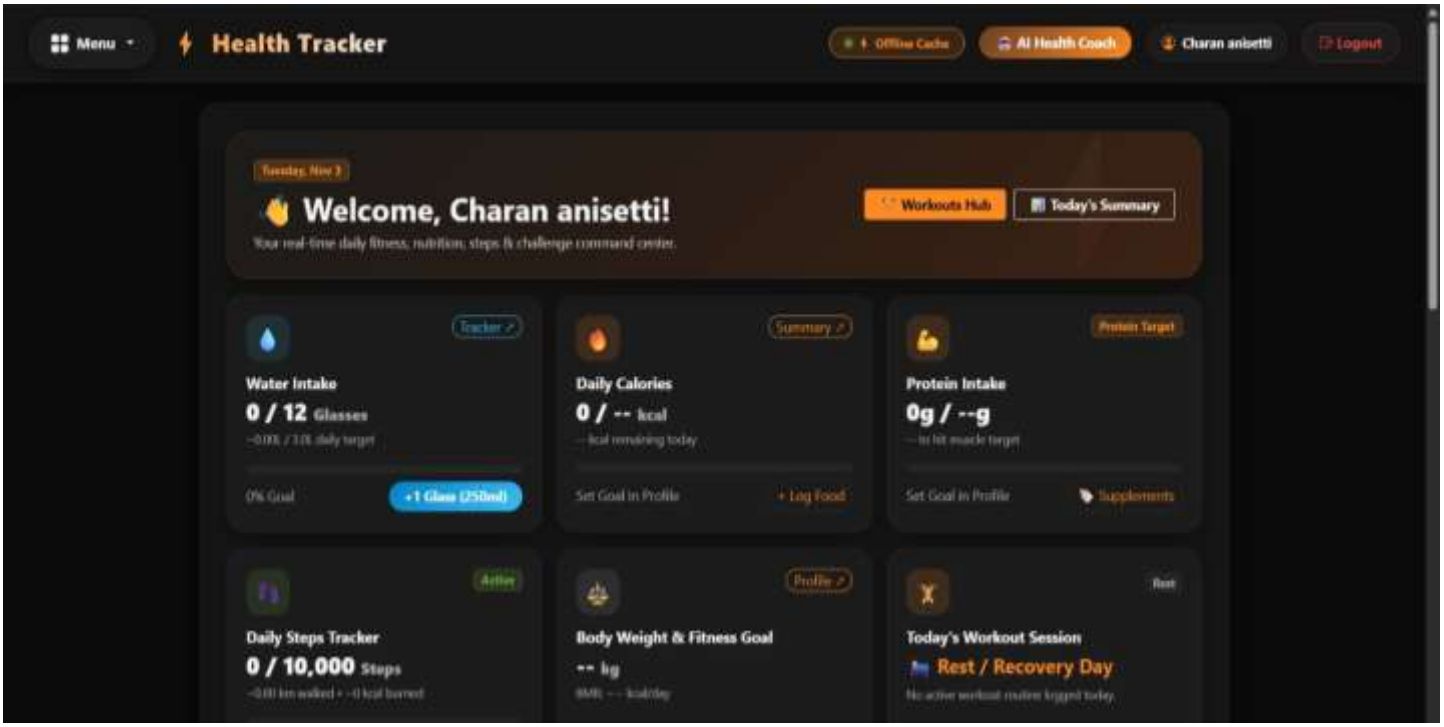


Figure 3: Daily Activity Tracking and 25-Day Transformation Challenge

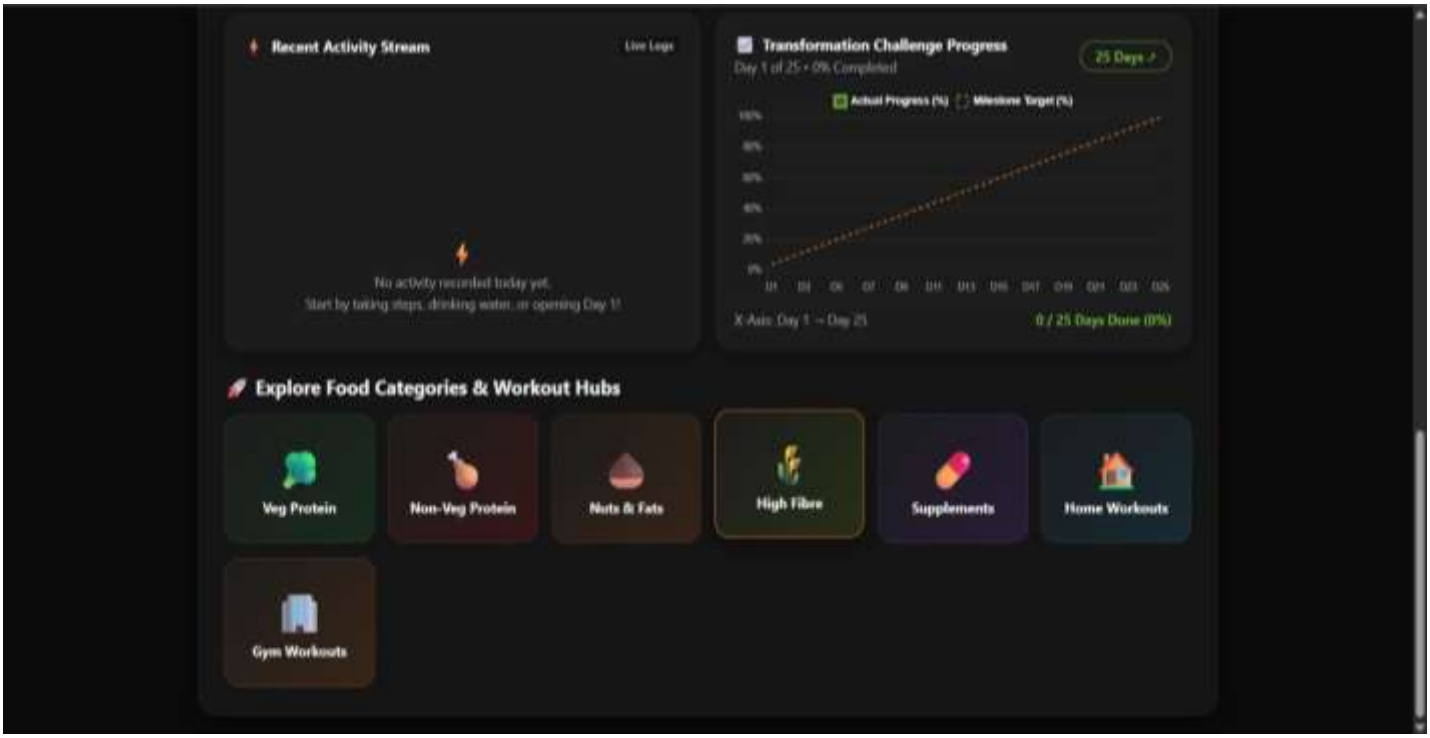


Figure 4: Calorie and Weight Gain Calculator Interface

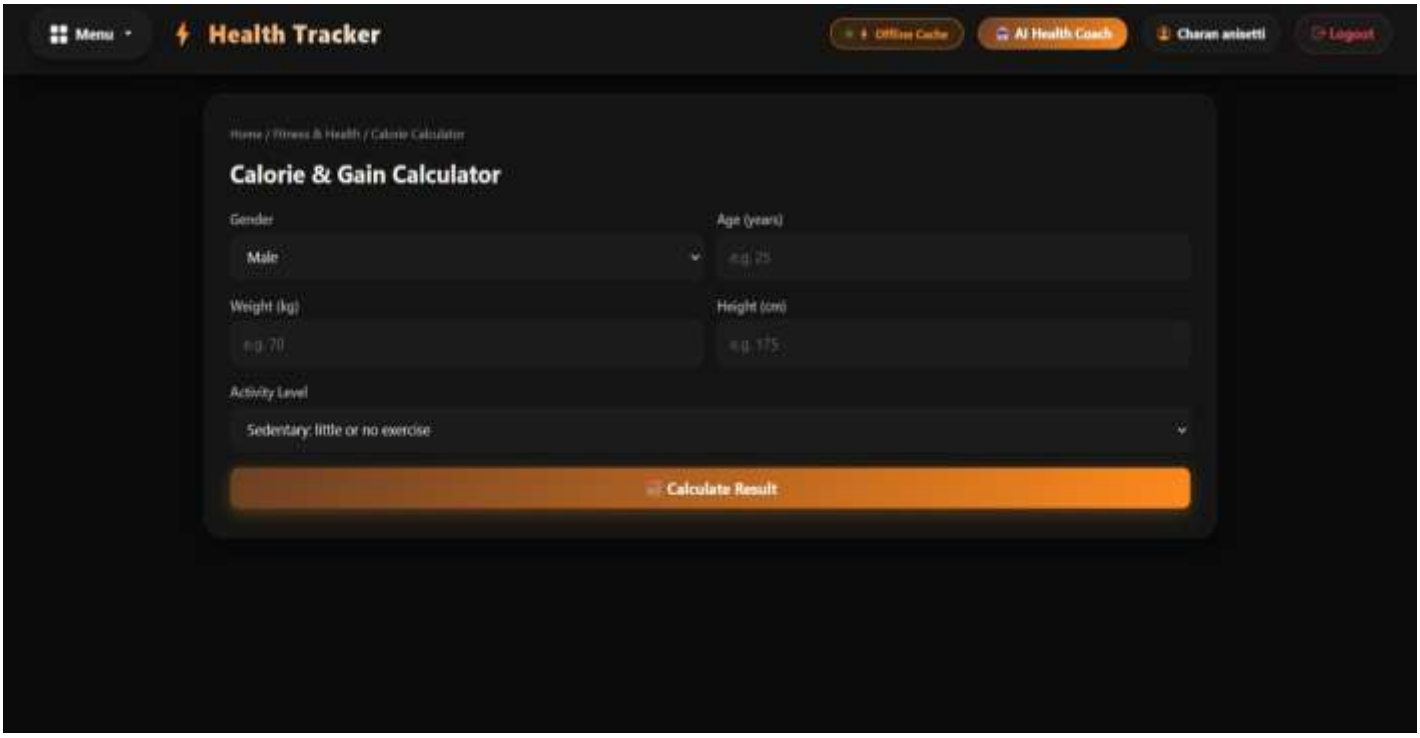


Figure 5: Calorie Calculation Results and Weight Gain Goals

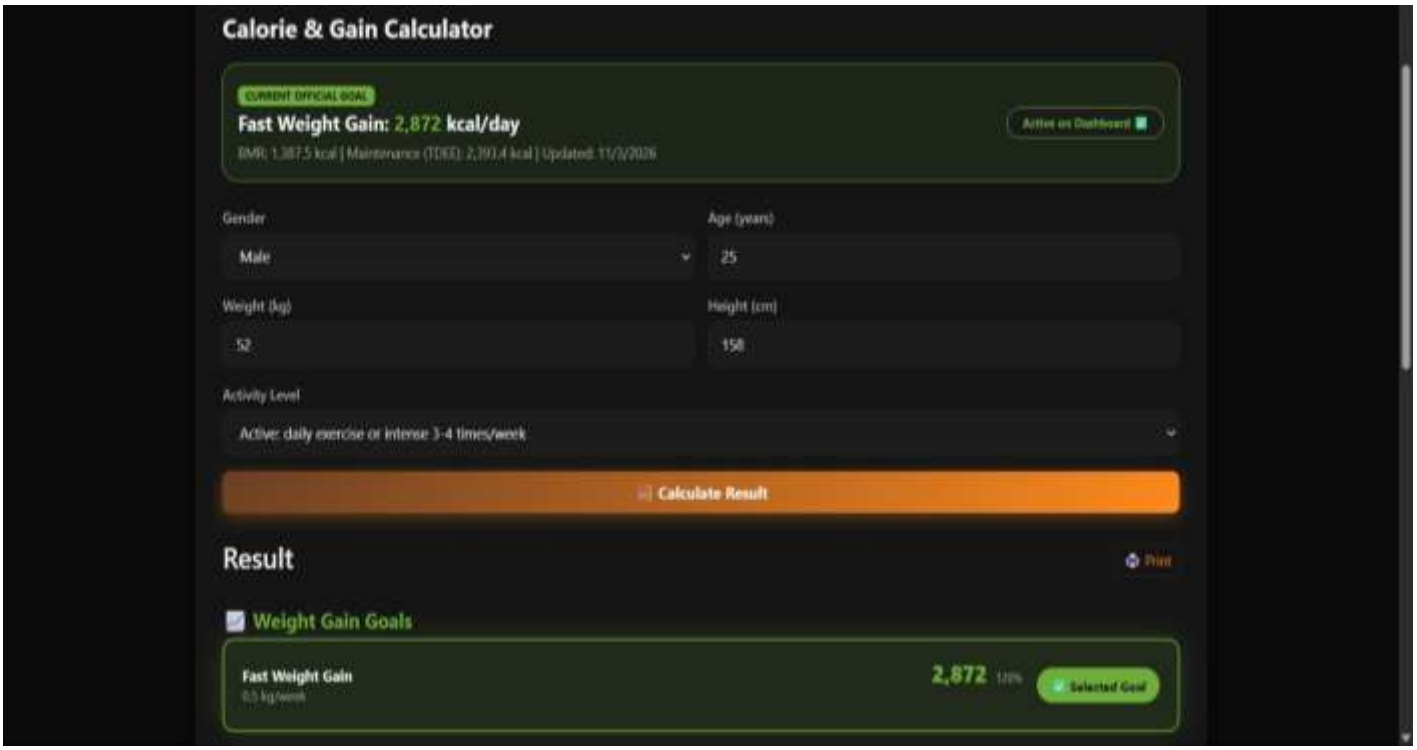


Figure 6: Workout Hub and Exercise Routine Selection

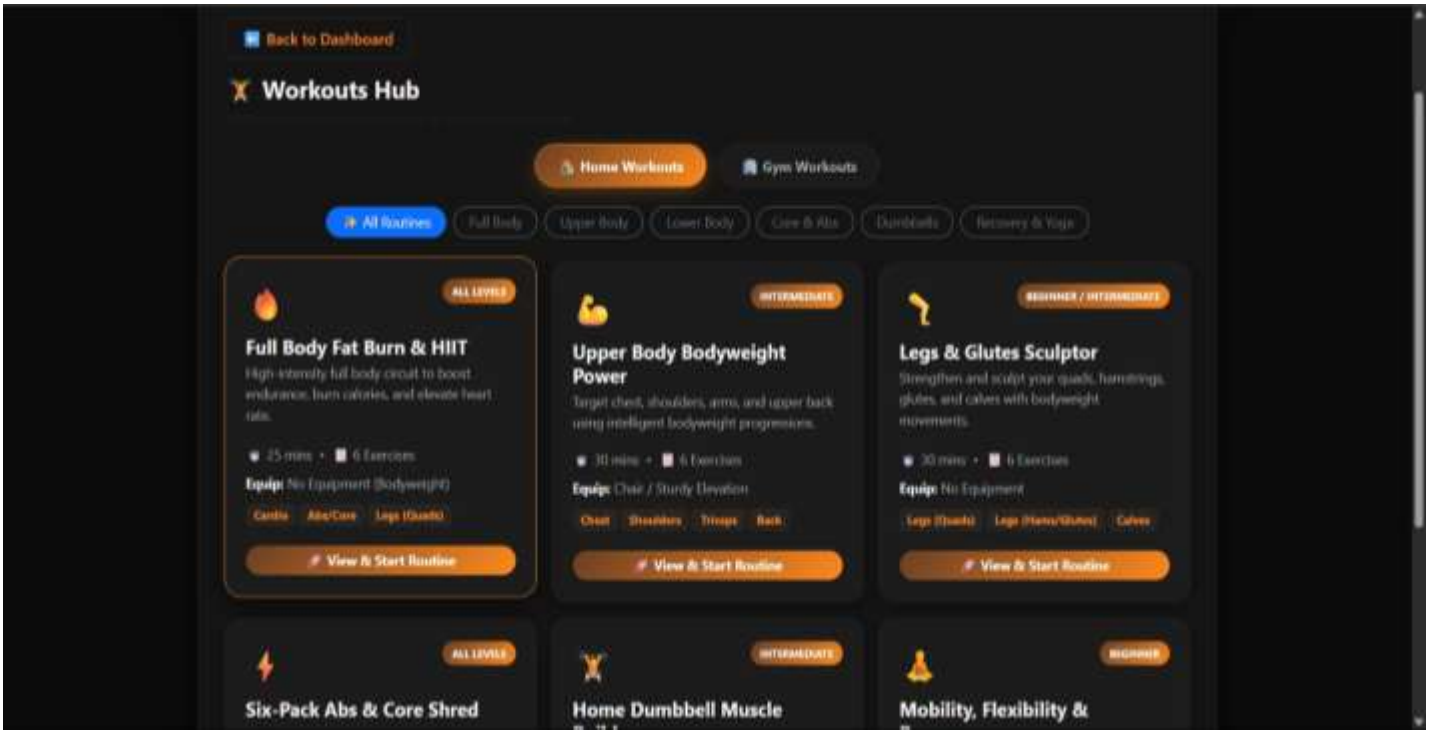


Figure 7: AI Health Coach Interface

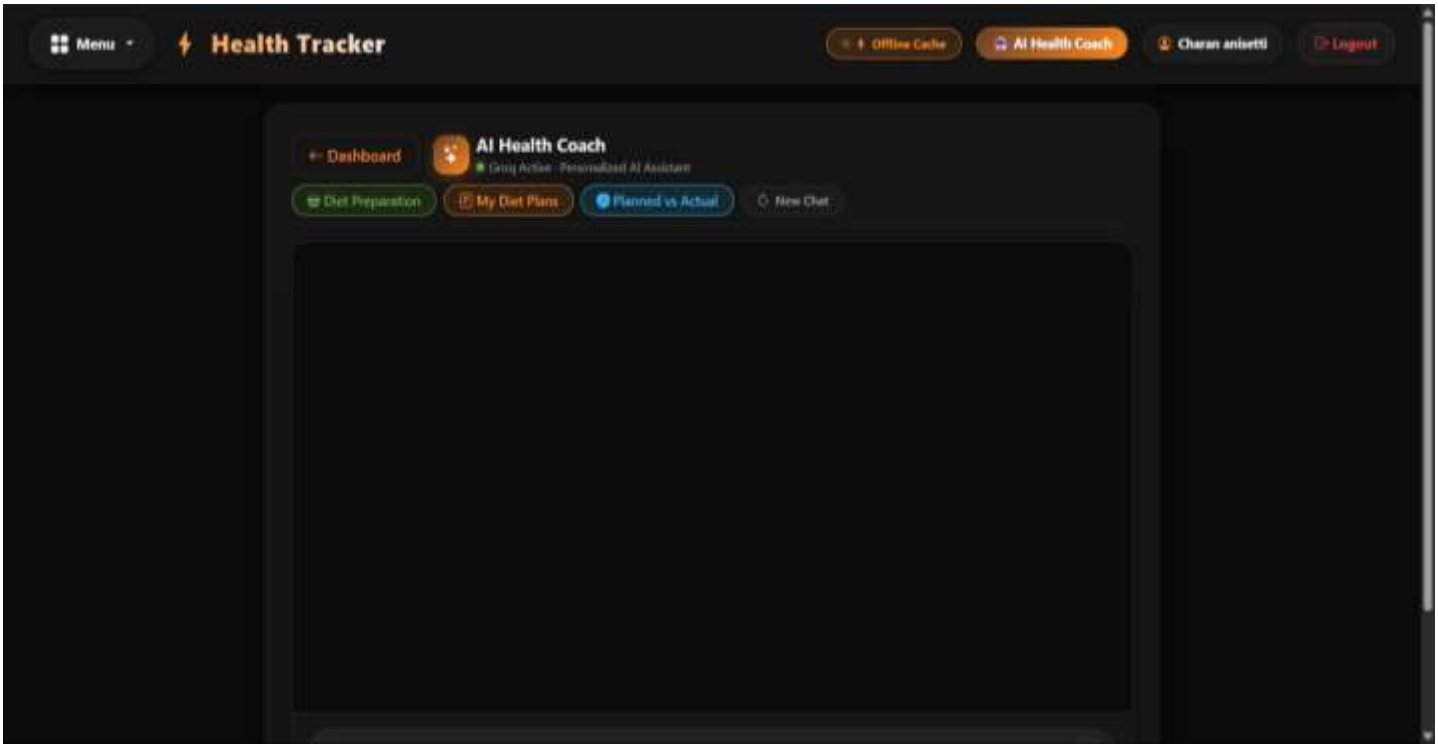


Figure 8: Confirmation of Available Food Items

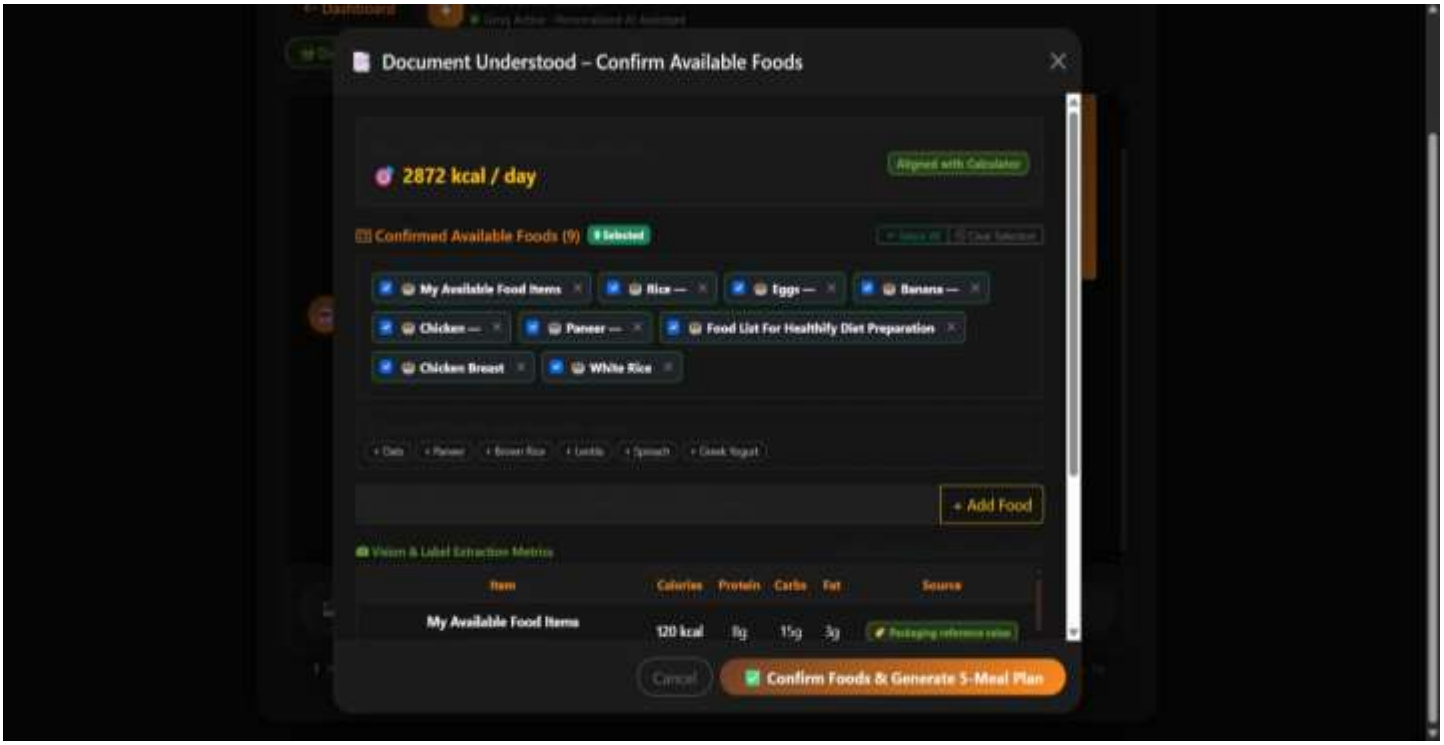


Figure 9: Selected Food Items for Diet Plan Generation

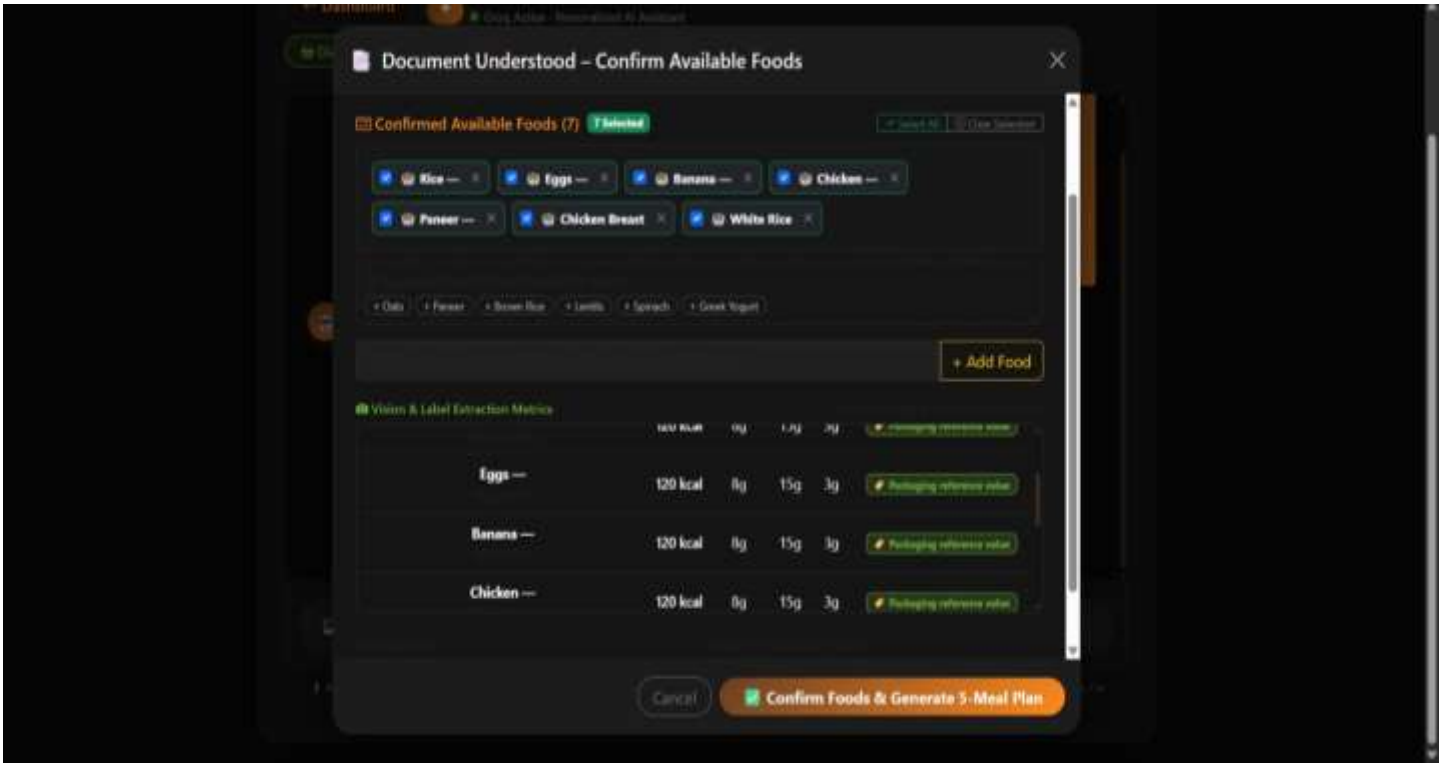


Figure 10: AI-Generated Daily 5-Meal Diet Plan



Figure 11: Main Health Tracker Dashboard

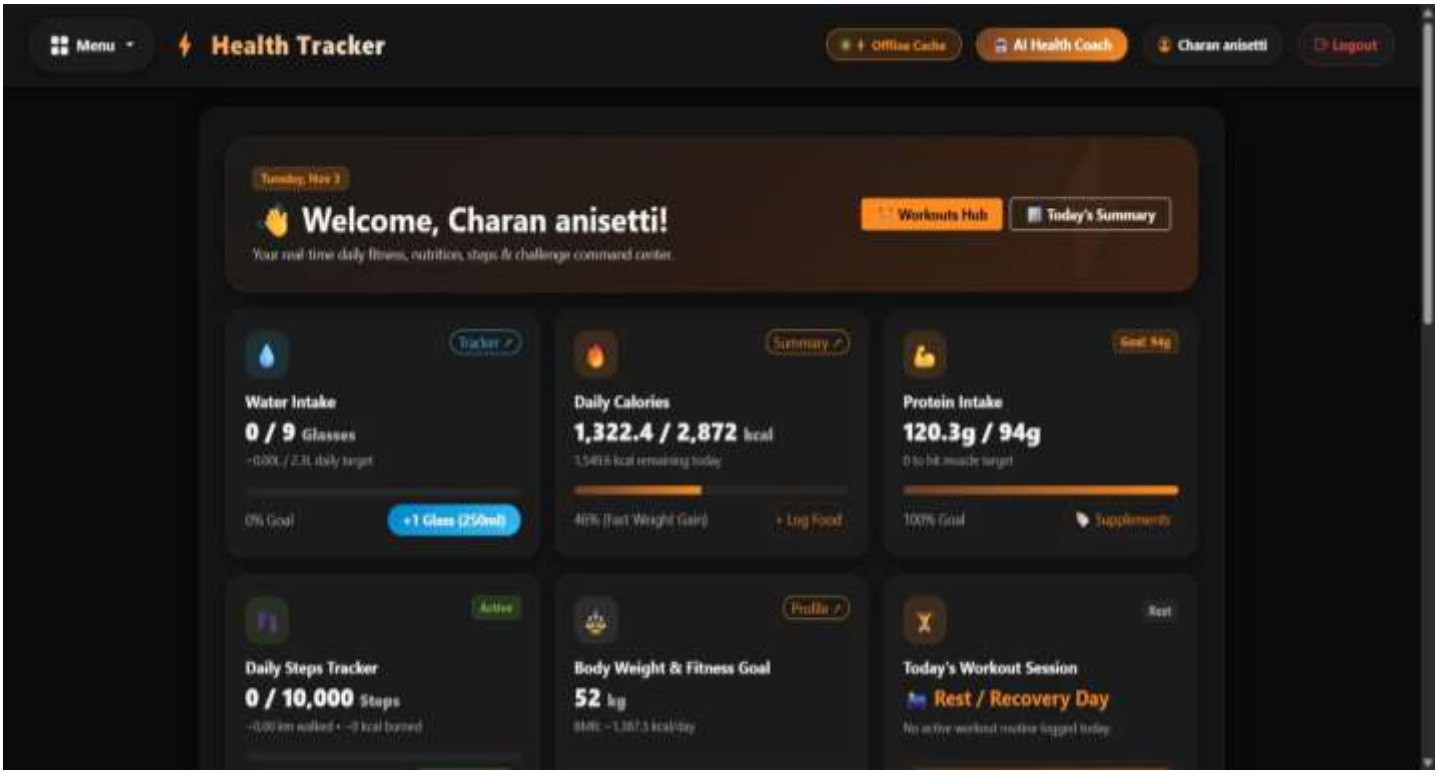


Figure 12: Daily Nutrition and Food Consumption Summary

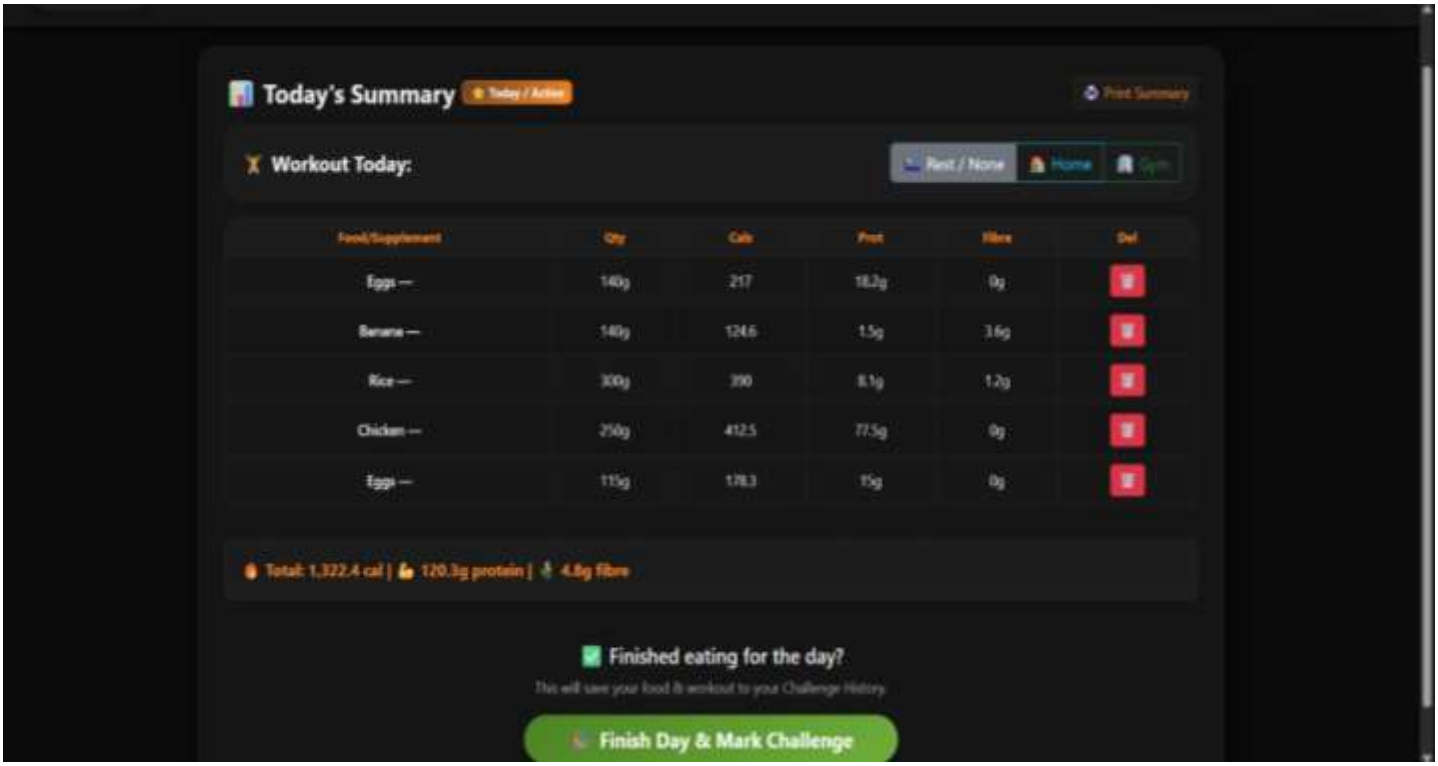


Figure 13: Daily Fitness Challenge Completion Status

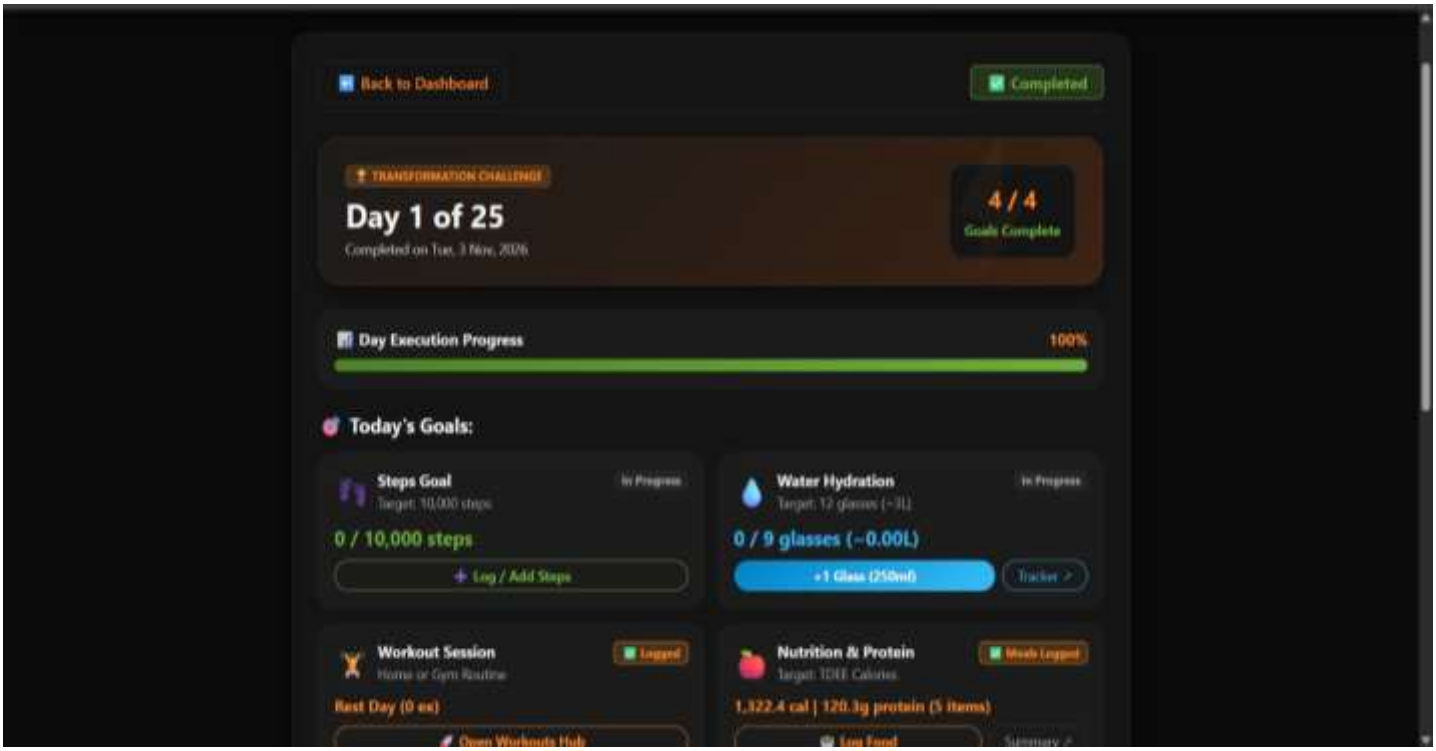
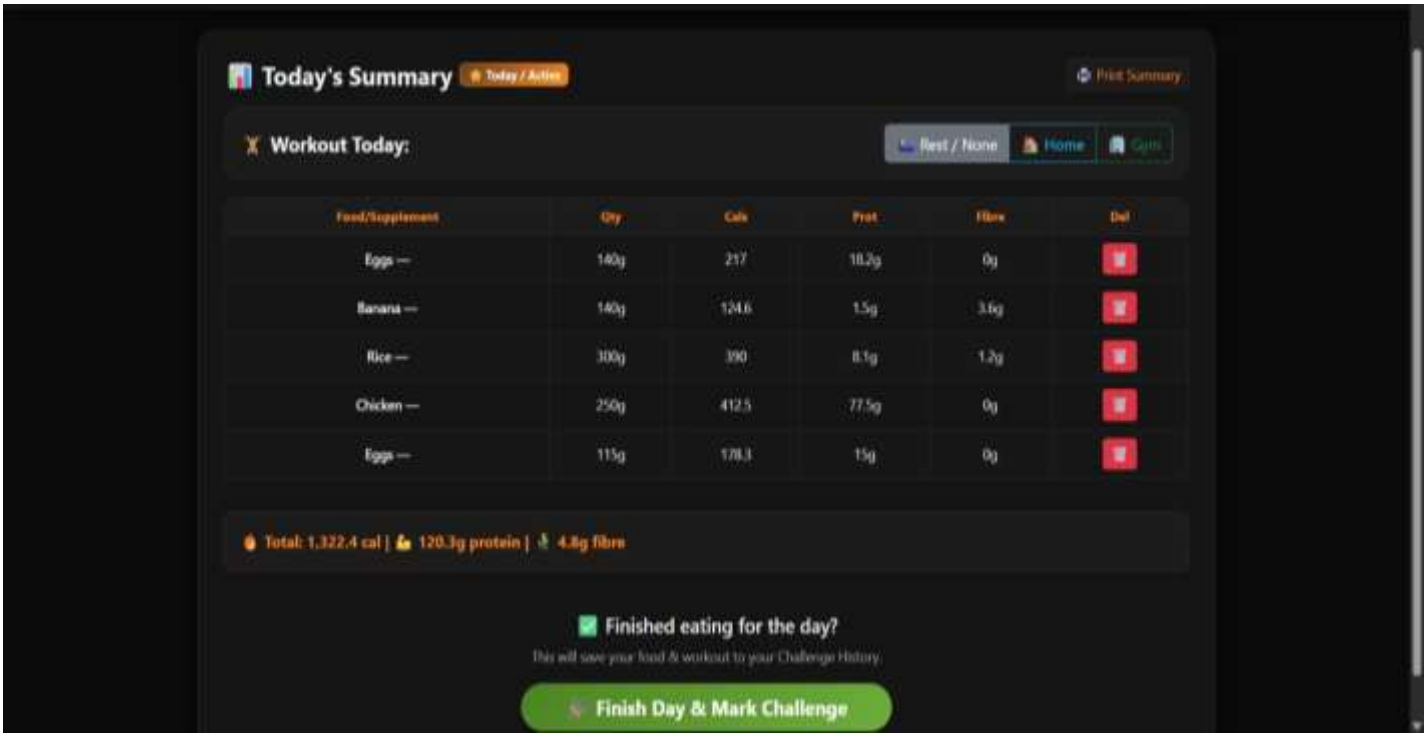


Figure 14: Detailed Food Consumption Report



VI. RESULTS AND TESTING

6.1 Results

The Healthify application provides an integrated platform for managing diet, nutrition, and fitness activities. It combines AI-based food recognition, personalized diet planning, and daily fitness tracking.

The system calculates daily calorie requirements based on user profile information and generates personalized diet plans. Users can upload food images, review detected food items, and monitor their nutritional intake.

The application also supports water intake tracking, step monitoring, workout logging, and a 25-day fitness challenge. MySQL provides persistent data storage, while IndexedDB supports selected offline functionality.

Overall, the application provides a convenient way for users to monitor their food consumption and fitness activities through a single platform.

6.2 Testing

The following testing activities are used to verify the application's functionality:

Test Case	Expected Result
User Registration	New users can create accounts.
User Login	Registered users can log in.
Profile Management	User profile details are saved and updated.
BMR Calculation	BMR is calculated using user details.
TDEE Calculation	Daily calorie requirements are calculated.
Food Image Recognition	Food items are identified from uploaded images.
Nutrition Retrieval	Nutritional information is displayed.
Diet Plan Generation	Personalized diet plans are generated.
Food Tracking	Food consumption records are saved.
Water Tracking	Daily water intake is recorded.
Step Tracking	Daily step information is maintained.
Workout Tracking	Workout activities are recorded.
Fitness Challenge	Challenge progress is maintained.
Database Integration	User and activity data are stored and retrieved.
Offline Storage	Supported data is available locally.
Data Synchronization	Stored changes are synchronized when connectivity returns.

6.3 Overall Outcome

The Healthify project combines Artificial Intelligence, web development, and database technologies to provide a smart diet and fitness management platform. It helps users monitor their nutrition, generate personalized diet plans, identify food items, and track daily fitness activities.

VII. CONCLUSION & FUTURE SCOPE

7.1 CONCLUSION

Healthify provides an AI-powered platform for managing diet, nutrition, and fitness activities. By integrating AI-based food recognition, nutritional information retrieval, personalized diet planning, and fitness tracking, the system helps users monitor their daily health-related activities.

The application calculates calorie requirements using BMR and TDEE, generates personalized diet plans, and supports food, water, step, and workout tracking. It also provides a 25-day fitness challenge to encourage consistent fitness habits.

The integration of Java Spring Boot, Node.js, MySQL, and GroqCloud AI services provides the technical foundation for the application. Offline storage and synchronization further improve accessibility.

Overall, Healthify aims to make diet planning and fitness tracking easier, more convenient, and more accessible through an integrated web application.

7.2 FUTURE SCOPE

- Improve food recognition accuracy using advanced AI models.
- Introduce real-time nutritional analysis.
- Add wearable device integration for automatic fitness tracking.
- Provide multilingual support for different users.
- Improve personalized diet recommendations based on user progress.
- Add advanced fitness reports and progress visualizations.
- Introduce voice-based food logging.
- Expand the nutrition database with additional food items.
- Add personalized workout recommendations.
- Improve offline functionality and data synchronization.
- Introduce additional fitness challenges and achievement systems.